

1. Intro/Preparation:

- Familiarize yourself with the function generator and the scope by demonstrating that you can **generate a waveform with a frequency of 10kHz and appropriate amplitude, and measure its amplitude, frequency, and phase on the oscilloscope**. You should be able to do this remotely in addition to manually using the controls on the front panel of the instruments. Use the code from the template scripts below to get started remotely.
- Both the scope and function generator are controllable via **template scripts** written in Python 3:
 - **scope_func_together.py**
 - **func_gen_control_template.py**
 - **scope_control_template.py**
 - **scope_func_measure.py**
- The scripts have been provided to you on the two raspberry Pi's at the **/usr/local/bin/skindepth** folder. Please copy them to your own directory on the Pi and edit them to accomplish the lab.
- In addition to the scripts, we have also included the (physical and software) manuals for both the function generator (Agilent/Hp 33120A) and the oscilloscope (RIGOL DS1022Z)
- Hints:
 - The task of setting, querying and measuring information remotely is very similar in sequence and logic to performing the same task manually. Use your intuition and the template scripts provided to accomplish this task.
 - The scripts **scope_func_together.py** and **scope_func_measure.py** include code stems for performing basic functions of the scope and function generator and is intended to reduce the complexity for you in terms of getting started. I would start here.

2. Setup:

The first time you log in, you will have to do this check in the terminal to make sure that everything is working properly.

```
~ python3
>> import pyvisa as visa
>> rm = visa.ResourceManager('@py')
>> dev_list = rm.list_resources()
>> fgen = rm.open\_resource(dev_list[0])
>> fgen.query("*IDN?")
>> exit()
```

3. Data Collection:

- We have provided the necessary theoretical background to convince you of the existence of the skin-depth effect (you should also have at some point derived it yourself). Now, **you must collect data which allows you to estimate the skin-depth of both aluminum and copper.** To aid you in this task, we have provided the following:
 - Two RF coils
 - 1 transmit (large diameter): A coil approximately 6" long, 2" in diameter, and wound with approximately 10 turns per cm
 - 1 receive (small diameter): A pick-up coil consisting of 200 turns of #38 Formex insulated copper wire wound on a 3/8" diameter form
 - Two metal tubes of appropriate diameter and thickness
 - 1 copper
 - 1 aluminum
 - Digital calipers
 - BNC connection hardware
 - if you need cables or connectors beyond those provided, ask the teaching team
 - Controllable function generator
 - Digital Oscilloscope
- A simple method for collecting suitable data is as follows:
 - Take data at frequencies between 100 Hz and 80 kHz, ensuring that the signal is not clipped or distorted. **Measure the amplitude and phase at each frequency for aluminum, copper, and air.** Take care in deciding how you will measure phase and define amplitude and what scripts you would like to use.
 - You may need to re-take some of the high-frequency pipe data with a higher-amplitude signal from the function generator if the signal devolves into noise. This is accomplished by taking multiple measurements of the same waveform to reduce noise. The amplitude and phase should be stable at each frequency, so you should be able to take measurement data robustly multiple times to improve signal to noise ratio. Do not exceed 10V peak-to-peak ($V_{pp} \leq 10$)

4. Data Analysis

- **Calculate the expected values of the amplitude ratio and phase difference at each frequency and compare them to their measured values.**
- **Perform non-linear least squares fitting of an appropriate model to your data to find the conductivities and compare to literature.** How different are your values from the accepted values?
- **For each metal, compare the results obtained by solving each data point for a skin depth value versus the expected results using your average conductivity found above and the expected conductivity.**

5. Extension

- You may have noticed that our suggested method for data collection is a bit brute-force. While it is robust, it is also time consuming and tedious.
- Look at the form of what you are measuring. Did you notice that you are effectively being asked to measure the transfer function of the system comprised of the tube and transmit and receive coils. Are there more efficient methods for measuring a transfer function?
- The function generator is programmable and capable of arbitrary waveform output, so feel free to read some of the suggested material below and **implement a more efficient method for data collection using the tools provided above**. Ask the teaching team if you need help with any of the theory, we would be happy to help you learn.
 - https://www.ece.rice.edu/~jdw/243_lab/file.38.html
 - https://en.wikipedia.org/wiki/Transfer_function#Gain,_transient_behavior_and_stability
 - <https://onlinelibrary.wiley.com/doi/10.1002/mrm.24263>
 - This is MRI specific however it contains a good derivation and method for transfer function measurement using triangular waveforms in the Theory and Methods section
 - https://thinksrs.com/downloads/pdfs/applicationnotes/SR1_SweptSine.pdf
 - This details how to use a sinusoidal sweep to measure the transfer function of an arbitrary system